Тогда

2015 | $\int_0^{\frac{1}{2}} \arcsin x\,dx$;

$$\left.\begin{array}{l} \int x^k \arcsin x\,dx \\ \int x^k \operatorname{arctg} x\,dx \\ \int x^k \ln x\,dx \end{array}\right\}$$ по части с выделением x^k под дифф.

По части:

$$\int_a^b f(x)\,d(g(x)) = f(x)g(x)\Big|_a^b - \int_a^b g(x)\,df(x)$$

$$\int_0^{\frac{1}{2}} \underbrace{(\arcsin x)}_{f}\,\underbrace{d(x)}_{g} = x\arcsin x\Big|_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x\,d(\arcsin x) =$$

$$= \frac{1}{2}\arcsin\frac{1}{2} - \int_0^{1/2} x\cdot\frac{1}{\sqrt{1-x^2}}\,dx =$$

$$= \frac{1}{2}\cdot\frac{\pi}{6} - \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\cdot x\,dx = \frac{\pi}{12} - \int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\,d\left(\frac{x^2}{2}\right)$$

Зачем $\left(\frac{x^2}{2}\right)' = x$, $\int x\,dx = \frac{x^2}{2}$

Бел

$$\int f(x)\,dg(x) = \int f(x)\cdot g'(x)\,dx$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$\arcsin\frac{1}{2} = \frac{\pi}{6}$$

Без. множител в подинтегралната ф-я влиза под диф. обс става примитивна, т.е.

$$\int f(x) \cdot h(x) dx = \int f(x) d H(x), \text{ където } H'(x) = h(x), \text{ т.е. } H \text{ е примитивна на } h$$

примитивна е обратното на производна

$$\left(\int h(x) dx = H(x) \right)$$

Изнася от диф. $d(.)$ обс става производна

Влиза в диф. $d(.)$ обс става примитивна

$\rightarrow$ от стр. 2 $= \frac{\pi}{12} - \int_0^{1/2} \frac{1}{\sqrt{1-x^2}} d\left(\frac{1}{2}\cdot x^2\right) = \frac{\pi}{12} - \frac{1}{2}\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} d(x^2) =$

$= \frac{\pi}{12} - \frac{1}{2}\cdot(-1)\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} d(1-x^2)$

$= \frac{\pi}{12} + \frac{1}{2}\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} d(1-x^2) =$

$y = 1 - x^2$

$\int \frac{1}{\sqrt{y}} dy =$

$= \int y^{-\frac{1}{2}} dy =$

$= \frac{y^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C = 2\sqrt{y} + C$

$\int f(x) d\, k g(x) =$

$= k \int f(x) d g(x)$

$\int f(x) d g(x) =$

$= \frac{1}{k} \int f(x) d(kg(x))$

$\int f(x)\, dg(x) = \int f(x) d(g(x)+C)$

Приложения Н.-Л. (неформално)

Нека $\int f(x)dx = F(x) + C$ (неопределен)

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

Кой задачата: $\int \frac{1}{\sqrt{1-x^2}} d(1-x^2) = 2\sqrt{1-x^2} + C \Rightarrow$ б.1.

$$= \frac{\pi}{12} + \frac{1}{2} \cdot 2\sqrt{1-x^2}\Big|_0^{\frac{1}{2}} = \frac{\pi}{12} + \left(\sqrt{\frac{3}{4}} - 1\right)$$

оно стр. 4

Ивн 2016 $\int_1^2 x \log_2 \frac{1}{x}\, dx =$

Бел.

$\log_a b = \frac{\ln b}{\ln a}$

$\log_2 \frac{1}{x} = \frac{\ln(\frac{1}{x})}{\ln 2} =$

$= -\frac{\ln x}{\ln 2}$, $\ln 1 = 0$

$= \int_1^2 -x\frac{\ln x}{\ln 2}\, dx = \frac{-1}{\ln 2} \underbrace{\int_1^2 x \ln x\, dx}_{I}$

$I = \int_1^2 x \ln x\, dx = \int_1^2 \ln x\, d\left(\frac{x^2}{2}\right) = \frac{x^2}{2}\cdot \ln x \Big|_1^2 - \int_1^2 \frac{x^2}{2}\, d(\ln x) =$

$= \frac{2^2}{2}\ln 2 - \int_1^2 \frac{x^2}{2}\, d(\ln x) \Rightarrow$

$$\rightarrow = 2\ln 2 - \int_1^2 \frac{x^2}{2} \cdot \frac{1}{x} dx = 2\ln 2 - \int_1^2 \frac{x}{2} dx$$

$$= 2\ln 2 - \frac{1}{2} \int_1^2 x\,dx = 2\ln 2 - \frac{1}{2} \left(\frac{x^2}{2} \Big|_1^2 \right) =$$

$$= 2\ln 2 - \frac{1}{2} \left(\frac{2^2}{2} - \frac{1^2}{2} \right) !$$

$(\ln x)' = \frac{1}{x}$

$\int \underline{\ln x}\, d\underline{x} =$

$= x\ln x - \int x\, d\ln x =$

$= x\ln x - \int \cancel{x} \cdot \frac{1}{\cancel{x}} dx = \underline{x\ln x - x}$

Сентябрь 2018

$$\int_0^1 x^2 \operatorname{arctg} x\,dx = \int_0^1 \operatorname{arctg} x\, d\left(\frac{x^3}{3}\right) =$$

$$= \operatorname{arctg} x \cdot \frac{x^3}{3}\Big|_0^1 - \int_0^1 \frac{x^3}{3}\, d(\operatorname{arctg} x) = \operatorname{arctg} 1 \cdot \frac{1}{3} -$$

$$- \frac{1}{3}\int_0^1 \frac{x^3}{x^2+1}dx = \frac{\pi}{12} - \frac{1}{3}\int_0^1 \frac{x^3}{x^2+1}dx = \longrightarrow$$

$$(\operatorname{arctg} x)' = \frac{1}{1+x^2}$$

$$\operatorname{arctg} 1 = \frac{\pi}{4}$$

$$\rightarrow = \frac{\pi}{12} - \frac{1}{3}\int_0^1 \frac{x^3+x-x}{x^2+1}\,dx = \frac{\pi}{12} - \frac{1}{3}\int_0^1 \frac{x(x^2+1)-x}{x^2+1}\,dx =$$

$$= \frac{\pi}{12} - \frac{1}{3}\int_0^1 \left(x - \frac{x}{x^2+1}\right)dx = \frac{\pi}{12} - \frac{1}{3}\left[\int_0^1 x\,dx - \int_0^1 \frac{x}{x^2+1}\,dx\right] =$$

$$= \frac{\pi}{12} - \frac{1}{3}\left[\frac{x^2}{2}\Big|_0^1 - \int_0^1 \frac{1}{x^2+1}\,d\left(\frac{x^2}{2}\right)\right] =$$

Бел $\int \frac{1}{y}\,dy = \ln|y| + C$

$$= \frac{\pi}{12} - \frac{1}{3}\left[\frac{1}{2} - \frac{1}{2}\int_0^1 \frac{1}{x^2+1}\,d\underbrace{(x^2+1)}_{y}\right] = \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6}\ln(x^2+1)\Big|_0^1 =$$

$$= \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6}\left|\ln 2 - \ln 1\right|$$

Септември 2019 | $\int_0^2 \ln(x^2+4)\,dx$, отколо по части

$$= x\ln(x^2+4)\Big|_0^2 - \int_0^2 x\,d\ln(x^2+4) = 2\ln 8 - \int_0^2 x\cdot\frac{1}{x^2+4}\cdot 2x\,dx$$

$$= 2\ln 8 - \int_0^2 \frac{2x^2}{x^2+4}dx = 2\ln 8 - 2\int_0^2 \frac{x^2+4-4}{x^2+4}dx =$$

$$= 2\ln 8 - 2\int_0^2 \left(1 - \frac{4}{x^2+4}\right)dx =$$

$$\left(\ln(x^2+4)\right)' = \frac{1}{x^2+4}\cdot(x^2+4)' = \frac{2x}{x^2+4}$$

$$= 2\ln 8 - 2\int_0^2 1\,dx - 2\int_0^2 \frac{-4}{x^2+4}\,dx = 2\ln 8 - 2x\Big|_0^2 + 8\int_0^2 \frac{1}{x^2+4}\,dx$$

$$= 2\ln 8 - 4 + 8\int_0^2 \frac{1}{x^2+4}\,dx =$$

$$= 2\ln 8 - 4 + 8 \cdot \frac{1}{2}\,\mathrm{arcctg}\left(\frac{x}{2}\right)\Big|_0^2$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\cdot \mathrm{arcctg}\left(\frac{x}{a}\right)$$

$$\mathrm{arcctg}\,1 = \frac{\pi}{4}$$

Септември 2016

$$\int_0^{\frac{\pi}{2}} x\cos^2 x\,dx =$$

$$\int x^k \sin ax\,dx$$

$$\int x^k \cos ax\,d$$

$$\int x^k e^{ax}\,dx$$

По части

Вкарай sin, cos, e, ln

диф., такова нещо, колкото

е степента на x, т.е. k

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\sin^2 x = \frac{1-\cos 2x}{2}$$

$$= \int_0^{\frac{\pi}{2}} x\left(\frac{1+\cos 2x}{2}\right) dx = \int_0^{\frac{\pi}{2}} \frac{x}{2} + \frac{x\cos 2x}{2}\,dx =$$

$$= \frac{x^2}{4}\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\underbrace{\int_0^{\frac{\pi}{2}} x\cos 2x\,dx}_{I} = \frac{\pi^2}{16} + \frac{1}{2}I$$

$$I = \int_0^{\frac{\pi}{2}} x\cos 2x\,dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} x\cos(2x)\,d(2x) =$$

$$\int \cos y\,dy = \sin y$$

$$\sin\pi = 0$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} x\,d\sin(2x) = \frac{1}{2}\left[x\sin(2x)\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin(2x)\,dx \right]$$

$$\int \sin y\,dy = -\cos y$$

$$= \frac{1}{2}\left(-\frac{1}{2}\int_0^{\frac{\pi}{2}} \sin(2x)\,d(2x) \right) = -\frac{1}{4}(-\cos 2x)\Big|_0^{\frac{\pi}{2}} = -\frac{1}{4}(1+1) = -\frac{1}{2}.$$

gdw wtw no racte (e^x neg grdo.)

$$I = \int e^x \sin x \, dx = \underbrace{\dots} - I$$

$$2I = \dots$$

$$\int x \sin x \cos(2x) \, dx =$$

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}$$

$$\int x \left(\frac{\sin 3x + \sin(-x)}{2} \right) dx = \int \frac{x \sin 3x}{2} dx + \int \frac{x \sin(-x)}{2} dx$$

Септември 2017

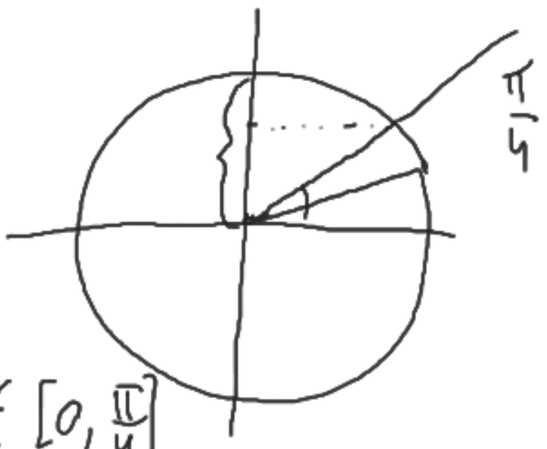
$$\int_0^{\pi/2} \left|\frac{1}{2} - \sin^2 x\right| dx$$

$$|y| = \begin{cases} y, & y \geq 0 \\ -y, & y < 0 \end{cases}$$

Интересуваме се за кои $x \in [0, \frac{\pi}{2}]$ $\frac{1}{2} - \sin^2 x \geq 0$ и кога е < 0

$\frac{1}{2} \geq \sin^2 x$, $\sin x \geq 0$ за $x \in [0, \frac{\pi}{2}]$, затова $\sqrt{\cdot}$

$\Leftrightarrow \frac{1}{\sqrt{2}} \geq \sin x$, равенство при $x = \frac{\pi}{4}$



$\Leftrightarrow \sin\frac{\pi}{4} \geq \sin x$ и тъй като $\sin$ расте в $[0, \frac{\pi}{2}]$, то н-вото е сила. за $x \in [0, \frac{\pi}{4}]$

$$\Rightarrow \quad \frac{1}{2} - \sin^2 x \begin{cases} \geq 0 & \text{за } x \in \left[0, \frac{\pi}{4}\right] \\ < 0 & \text{за } x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right] \end{cases} \Rightarrow$$

$$\left| \frac{1}{2} - \sin^2 x \right| = \begin{cases} \frac{1}{2} - \sin^2 x, & \text{за } x \in \left[0, \frac{\pi}{4}\right] \\ -\frac{1}{2} + \sin^2 x, & \text{за } x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \end{cases}$$

Като интеграл

$$\int_0^{\frac{\pi}{2}} \left| \frac{1}{2} - \sin^2 x \right| dx = \int_0^{\frac{\pi}{4}} \left| \frac{1}{2} - \sin^2 x \right| dx + \int_{\frac{\pi}{4}}^{\pi/2} \left| \frac{1}{2} - \sin^2 x \right| dx =$$

$$= \underbrace{\int_0^{\frac{\pi}{4}} \left(\frac{1}{2} - \sin^2 x\right) dx}_{I_1} + \underbrace{\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(-\frac{1}{2} + \sin^2 x\right) dx}_{I_2 - \text{за вибяги}} =$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$I_1 = \int_0^{\frac{\pi}{4}} \frac{1}{2} - \sin^2 x \, dx = \int_0^{\frac{\pi}{4}} \frac{1}{2} - \frac{1 - \cos 2x}{2} \, dx = \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos 2x \, dx =$$

$$= \frac{1}{2}\cdot\frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2x \, d(2x) = \frac{1}{4} \sin 2x \Big|_0^{\frac{\pi}{4}} = \frac{1}{4}.$$